

Sejin Kim

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EDUCATION	Ph.D. in School of Computing, KAIST (<i>Advisor: Jae-Gil Lee</i>)	Feb 2017 – Feb 2023
	▪ Thesis: Region-Level Spatio-Temporal Prediction Based on Transfer Learning for Data-Poor Cities	
	M.S. in School of Computing, KAIST (<i>Advisor: Kyu-Young Whang</i>)	Sep 2013 – Aug 2016
	▪ Thesis: Design and Implementation of Cost-Based Query Optimization for Odysseus/OOSQL	
	B.S. in Department of Computer Science, KAIST	Feb 2009 – Aug 2013
RESEARCH PROFILE	Human-like Reasoning and ARC-AGI ▪ Reasoning Analysis and Benchmarks: [J03], [C17], [C08], [C07], [DJ01], [DC07] ▪ Environments, Platforms, and Human Trajectory Data: [C04], [C06], [C10], [DC09] Program Synthesis and Neuro-Symbolic Reasoning ▪ LLM-Guided, Critic-Guided, and Latent Program Synthesis: [C18], [C15], [C13], [C12] ▪ Object-Centric, Analogical, and Model-Based Reasoning: [C05], [C03], [DC11], [DC10], [DC08] Reinforcement Learning and Decision-Making ▪ Offline RL, Diffusion-Based RL, and Environment Design: [C16], [C14], [C11] ▪ Probabilistic Exploration and Solution Augmentation: [J02] Human-AI Alignment and Applied AI ▪ Human-AI Alignment and Misalignment Analysis: [C09] ▪ User Modeling and Social Computing: [C02], [C01], [DC02] ▪ Applied Machine Learning and Systems: [J01], [DC06], [DC05], [DC04], [DC03], [DC01]	
WORK EXPERIENCE	Postdoctoral Researcher – Data Science Lab, GIST	Apr 2023 – Aug 2026
	▪ Conducted research on human-like reasoning, reinforcement learning, and program synthesis.	
	Researcher – GraphAI Co., Ltd.	Feb 2023 – Mar 2023
	▪ Developed core technologies for GraphDB.	
	Researcher – Advanced Information Technology Research Center, KAIST	Sep 2016 – Feb 2017
	▪ Contributed to the development and release of Odysseus/OOSQL.	
TEACHING EXPERIENCE	Lecturer ▪ Intensive Python Course (40h), Kongju National University ▪ Korea Olympiad in Informatics (KOI), WITLab	
	Teaching Assistant, KAIST ▪ CS101 Introduction to Programming, School of Computing ▪ CS204 Discrete Mathematics, School of Computing ▪ CS206 Data Structures, School of Computing ▪ CS220 Programming Principles, School of Computing ▪ CS360 Introduction to Databases, School of Computing ▪ CS560 Database Systems, School of Computing ▪ CS664 Advanced Database Systems, School of Computing ▪ KSE525 Data Mining and Knowledge Discovery, Department of Knowledge Service Engineering ▪ KSE526 Analytical Methodologies for Big Data, Department of Knowledge Service Engineering ▪ KSE625 Data Mining for Social Networks, Department of Knowledge Service Engineering	
		Feb 2026 Mar 2017 – Feb 2023

PUBLICATIONS (*: co-first author, †: co-corresponding author)

- [C18] Jinwoo Jeon, Seong-Woong Shim, **Sejin Kim**[†], Sundong Kim[†], Byung-Jun Lee[†], “Neural MCTS with LLM Guidance for Effective Program Synthesis on Abstraction and Reasoning Corpus”.
ECML-PKDD, 2026 (*Journal Track; to appear in Machine Learning (SCIE, IF: 4.9, JCR Q2)*)
https://sejinkimm.github.io/files/ECML_PKDD_2026.pdf
- [C17] **Sejin Kim**, Hayan Choi, Seokki Lee, Sundong Kim, “ARCTraj: A Dataset and Benchmark of Human Reasoning Trajectories for Abstract Problem Solving”.
KDD, 2026 (*Acceptance rate: 21%*)
<https://dl.acm.org/doi/10.1145/3770854.3785683>
- [C16] Yunho Kim, Jaehyun Park, Heejun Kim, **Sejin Kim**[†], Byung-Jun Lee[†], Sundong Kim[†], “On the Role of Proposal Support in Diffusion-Based Offline RL for Sequential Decision-Making”.
ICML, 2026 (*Workshop on Decision-Making from Offline Datasets to Online Adaptation*).
<https://openreview.net/forum?id=roLdW7qeCO>
- [C15] Woletemariam Liyew, Dojun Oh, Seokki Lee, **Sejin Kim**, and Sundong Kim, “Semantic Mutation via LLMs: A Hybrid Approach to Evolutionary Program Synthesis”.
GECCO, 2026 (*Companion*).
https://sejinkimm.github.io/files/GECCO_2026_Companion.pdf
- [C14] Geonwoo Cho, Jaegyun Im, Jihwan Lee, Hojun Yi, **Sejin Kim**[†], Sundong Kim[†], “TRACED: Transition-aware Regret Approximation with Co-learnability for Environment Design”.
ICLR, 2026 (*Acceptance rate: 27%*).
<https://iclr.cc/virtual/2026/10010241>
- [C13] Marha Midhaty Rusli, Donghyeon Shin, **Sejin Kim**, Sundong Kim, “Tracing and Correcting Programs: Critic-Guided Synthesis for Visual Reasoning”.
AAAI, 2026 (*Bridge Program on Logical and Symbolic Reasoning in Language Models*).
<https://openreview.net/forum?id=6sB1rHJ1BN>
- [C12] Jumyung Park, Jiwon Park, Jinseo Shim, **Sejin Kim**, Paulina Vennemann, Sundong Kim, “Composed Program Induction with Latent Program Lattice”.
NeurIPS, 2025 (*Workshop on Symmetry and Geometry in Neural Representations*).
<https://neurips.cc/virtual/2025/136882>
- [C11] Jaehyun Park, Yunho Kim, **Sejin Kim**, Byung-Jun Lee, Sundong Kim, “Diffusion-model-guided Implicit Q-learning with Adaptive Revaluation”.
CoRL, 2025 (*Workshop on Learning Effective Abstractions for Planning*).
<https://openreview.net/forum?id=zIq6SuFooZ>
- [J03] Seungpil Lee, Woonchang Sim, Donghyeon Shin, Wongyu Seo, Jiwon Park, Seokki Lee, Sanha Hwang, **Sejin Kim**, Sundong Kim, “Reasoning Abilities of Large Language Models: In-Depth Analysis on the Abstraction and Reasoning Corpus”.
ACM Transactions on Intelligent Systems and Technology, 16(6), 2025 (*SCIE, IF: 6.6, JCR Q1*).
<https://dl.acm.org/doi/full/10.1145/3712701>
- [J02] Sanha Hwang, Seungpil Lee, **Sejin Kim**[†], Sundong Kim[†], “Solution Augmentation for ARC-AGI Problems Using GFlowNet: A Probabilistic Exploration Approach”.
Transactions on Machine Learning Research, 2025 (*Scopus*).
https://tmlr.infinite-conf.org/paper_pages/ULCOhBgGzy.html
- [C10] **Sejin Kim**, Hosung Lee, Sundong Kim, “Addressing and Visualizing Misalignments in Human Task-Solving Trajectories”.
KDD, 2025 (*Acceptance rate: 18%*).
<https://dl.acm.org/doi/10.1145/3711896.3736831>
- [C09] **Sejin Kim**, Hosung Lee, Sundong Kim, “Addressing and Visualizing Misalignments in Human Task-Solving Trajectories”.
ICLR, 2025 (*Workshop on Bidirectional Human-AI Alignment*).
<https://iclr.cc/virtual/2025/34288>
- [C08] **Sejin Kim**, Sundong Kim, “System-2 Reasoning via Generality and Adaptation”.
NeurIPS, 2024 (*Workshop on System-2 Reasoning At Scale*).
<https://neurips.cc/virtual/2024/104297>

- [C07] Seungpil Lee, Woochang Sim, Donghyeon Shin, **Sejin Kim**, Sundong Kim, “Reasoning Abilities of Large Language Models through the Lens of Abstraction and Reasoning”.
NeurIPS, 2024 (*Workshop on System-2 Reasoning At Scale*).
<https://neurips.cc/virtual/2024/104295>
- [C06] Suyeon Shim, Dohyun Ko, Hosung Lee, Seokki Lee, Doyoon Song, Sanha Hwang, **Sejin Kim**, Sundong Kim, “O2ARC 3.0: A Platform for Solving and Creating ARC Tasks”.
IJCAI, 2024 (*Demo*).
<https://doi.org/10.24963/ijcai.2024/1034>
- [C05] Jihwan Lee, Woochang Sim, **Sejin Kim**[†], Sundong Kim[†], “Enhancing Analogical Reasoning in the Abstraction and Reasoning Corpus via Model-Based RL”.
IJCAI, 2024 (*Workshop on the Interactions between Analogical Reasoning and Machine Learning*).
https://iarml2024-ijcai.loria.fr/files/2024/07/WorldModel_IJCAIW2024.pdf
- [C04] Hosung Lee*, **Sejin Kim***, Seungpil Lee, Sanha Hwang, Jihwan Lee, Byung-Jun Lee, Sundong Kim, “ARCLE: The Abstraction and Reasoning Corpus Learning Environment for Reinforcement Learning”.
CoLLAs, 2024 (*Acceptance rate: 25%*).
<https://proceedings.mlr.press/v274/lee25a.html>
- [C03] Jaehyun Park, Jaegyun Im, Sanha Hwang, Mintaek Lim, Sabina Ualibekova, **Sejin Kim**, Sundong Kim, “Unraveling the ARC Puzzle: Mimicking Human Solutions with Object-Centric Decision Transformer”.
ICML, 2023 (*Workshop on Interactive Learning with Implicit Human Feedback*).
<https://openreview.net/forum?id=F9QfmL6ljZ>
- [C02] Sundong Kim, Hwanjun Song, **Sejin Kim**, Beomyoung Kim, Jae-Gil Lee, “Revisit Prediction by Deep Survival Analysis”.
PAKDD, 2020 (*Acceptance rate: 21%*).
https://doi.org/10.1007/978-3-030-47436-2_39
- [J01] Kyu-Young Whang, Ijnu Na, Tae-Seob Yun, Jin-Ah Park, Kyu-Hyun Cho, **Sejin Kim**, Ilyeop Yi, Byung Suk Lee, “Building Social Networking Services Systems using the Relational Shared-Nothing Parallel DBMS”.
Data & Knowledge Engineering, 125, 101756, 2020 (*SCIE, IF: 3.9, JCR Q2*).
<https://doi.org/10.1016/J.DATAK.2019.101756>
- [C01] Sang-Jun Yea, **Sejin Kim**, John-Michaël TO, Jae-Gil Lee, “SentiWorld: Understanding Emotions between Countries Based on Tweets”.
ICWSM, 2016 (*Demo*).
<https://doi.org/10.1609/ICWSM.v10i1.14706>

DOMESTIC (*: co-first author, †: co-corresponding author)

- PUBLICATIONS**
- [DJ01] Woochang Sim, Hyebin Jin, **Sejin Kim**, Sundong Kim, “The Possibility of Prompt Engineering for ARC Problem Solving”.
KIISE Transactions on Computing Practices 30(2), 2024.
<https://doi.org/10.5626/KTCP.2024.30.2.063>
 - [DC11] Jihwan Lee, Seungpil Lee, **Sejin Kim**[†], Sundong Kim[†], “Extracting the core knowledge of ARC with the World Model”.
Korea Software Congress, 2023.
<https://www.dbpia.co.kr/journal/articleDetail?nodeId=NODE11705335>
 - [DC10] Gyojoon Gu, Woochang Sim, Jaegyun Im, **Sejin Kim**, Sundong Kim, “Using Contrastive Learning for Abstraction and Reasoning task”.
Korea Software Congress, 2023.
<https://www.dbpia.co.kr/journal/articleDetail?nodeId=NODE11705255>
 - [DC09] Hosung Lee*, **Sejin Kim***, Sundong Kim, “ARCLE: Gymnasium Environment for Abstraction and Reasoning Corpus”.
Korea Software Congress, 2023.
<https://www.dbpia.co.kr/journal/articleDetail?nodeId=NODE11705128>

- [DC08] Jaehyun Park, Jaegyun Im, Youngdo Lee, Donghyeon Shin, **Sejin Kim**, Sundong Kim, “Abstraction and Reasoning Challenge with Decision Transformer”.
Korea Computer Congress, 2023.
<https://www.dbpia.co.kr/journal/articleDetail?nodeId=NODE11488083>
- [DC07] Woochang Sim, Hyebin Jin, **Sejin Kim**, Sundong Kim, “The Possibility of Prompt Engineering for ARC Problem Solving”.
Korea Computer Congress, 2023.
<https://www.dbpia.co.kr/journal/articleDetail?nodeId=NODE11488050>
- [DC06] Yooju Shin, Dongmin Park, Junhyeok Kang, **Sejin Kim**, “Contrastive Learning with Segment Supervision for Semi-Supervised Time Series Classification”.
Korea Software Congress, 2022.
<https://www.dbpia.co.kr/journal/articleDetail?nodeId=NODE11224263>
- [DC05] Byunghyun Kim, Jihye Na, Yeongsin Byeon, **Sejin Kim**, Jae-Gil Lee, “Autoencoder ODEs for Time Series Anomaly Detection”.
Korea Software Congress, 2022.
<https://www.dbpia.co.kr/journal/articleDetail?nodeId=NODE11224165>
- [DC04] Joeun Kim, Doyoung Kim, **Sejin Kim**, Junhyeok Kang, Dongmin Park, Minseok Kim, Youngjun Lee, Jae-Gil Lee, “Performance Comparison of Self-Driving Models With and Without LIDAR Sensors Based on Reinforcement Learning in AWS DeepRacer Platform”.
Korea Computer Congress, 2022.
<https://www.dbpia.co.kr/journal/articleDetail?nodeId=NODE11113594>
- [DC03] Patara Trirat, Yooju Shin, **Sejin Kim**[†], Minseok Kim[†], “Generating a Machine Learning Model with a Few Sentences”.
Korea Software Congress, 2021.
<https://www.dbpia.co.kr/journal/articleDetail?nodeId=NODE11035819>
- [DC02] Beomyoung Kim, Sundong Kim, **Sejin Kim**, Jae-Gil Lee, “Customer Revisit Prediction Using Macroscale Mobility Information”.
Korea Software Congress, 2019.
<https://www.dbpia.co.kr/journal/articleDetail?nodeId=NODE09301908>
- [DC01] **Sejin Kim**, “A Comparison of Implementation Method of Spatial Operator in Spatial OLAP”.
Korea Computer Congress, 2018.
<https://www.dbpia.co.kr/journal/articleDetail?nodeId=NODE07502958>

PATENTS

- [P06] Geonwoo Cho, Jaegyun Im, **Sejin Kim**, Sundong Kim, “METHODS AND SYSTEMS FOR LEARNING BASED ON DIFFICULTY AND MUTUAL LEARNING”.
Korean Patent Application No: 10-2025-0160737, Oct 30, 2025.
- [P05] Jaehyun Park, Sundong Kim, **Sejin Kim**, Yunho Kim, “METHOD FOR TRAINING AN ADAPTIVE RE-EVALUATION-BASED EXECUTION SYSTEM”.
Korean Patent Application No: 10-2025-0148979, Oct 15, 2025.
- [P04] Yunho Kim, Sundong Kim, **Sejin Kim**, Heejun Kim, Jaehyun Park, “METHOD AND SYSTEM FOR RULE-BASED SYNTHETIC DATA GENERATION OFFLINE REINFORCEMENT LEARNING”.
Korean Patent Application No: 10-2025-0148978, Oct 15, 2025.
- [P03] **Sejin Kim**, Hosung Lee, Sundong Kim, “METHOD AND SYSTEM FOR AN INTENTION ESTIMATION IN HUMAN PROBLEM-SOLVING TRAJECTORIES”.
Korean Patent Application No: 10-2025-0135338, Sep 19, 2025.
- [P02] Hosung Lee, **Sejin Kim**, Sundong Kim, “METHOD AND SYSTEM FOR IMPLEMENTING A REINFORCEMENT LEARNING ENVIRONMENT CAPABLE OF SOLVING ABSTRACTION AND REASONING CHALLENGE”.
Korean Patent Application No: 10-2024-0101165, Jul 30, 2024.
<https://doi.org/10.8080/1020240101165>
- [P01] Jae-Gil Lee, Chunhee Lee, Jungeun Kim, **Sejin Kim**, “RETRIEVAL SYSTEM FOR GEOSOCIAL NETWORKS AND METHOD OF THE SAME”.
Korean Patent Registration No: 10-22055900, Dec 09, 2019.
<https://doi.org/10.8080/1020180029244>

GRANTS	R&D Innovation Program, funded by KISTI	Jan 2026 – Dec 2026
	▪ Title: Representation Learning for Intention-Driven Generalization and Adaptation	
	▪ GPU resources (Neuron system, 10 nodes)	
	Postdoctoral Fellowship, funded by NRF	Sep 2024 – Aug 2026
	▪ Title: Developing Abstraction and Reasoning Capabilities for AI Models	
	▪ KRW 120 million for 2 years.	
	AI-Based Convergence Talent Training Support Project, funded by GIST	Apr 2023 – Dec 2024
	▪ Title: Foundational AI Techniques on Abstraction and Reasoning	
	▪ KRW 60 million for 2 years.	
INVITED TALKS	▪ ChemEng Graduate Seminar, Sunchon National University (SCNU)	Jul 2026
	▪ Tutorial Session, Korea Computer Congress (KCC)	Jun 2026
	▪ Special Lecture for Process Optimization, Kongju National University (KNU)	Apr 2026
	▪ Early-Career Researcher Session, Korea Software Congress (KSC)	Dec 2025
	▪ Special Lecture for Generative AI, Sunchon National University (SCNU)	Nov 2025
	▪ AI Colloquium, Gwangju Institute of Science and Technology (GIST)	Oct 2025
	▪ CS Graduate Seminar, Ulsan National Institute of Science and Technology (UNIST)	Sep 2025
	▪ AXEL Camp, Gwangju Institute of Science and Technology (GIST)	Aug 2025
	▪ DIDCS Workshop, Korea Disease Control and Prevention Agency (KDCA)	May 2025
	▪ EECE Seminar, University of British Columbia (UBC)	Dec 2024
SERVICES	Area Chair	
	▪ ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD) 2026.	
	Session Chair	
	▪ ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD) 2025–2026.	
	Program Committee Member	
	▪ International Conference on Machine Learning (ICML) 2026.	
	▪ ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD) 2025–2026.	
	▪ International Conference on Learning Representations (ICLR) 2025.	
	▪ Conference on Neural Information Processing Systems (NeurIPS) 2024–2026.	
	▪ Association for the Advancement of Artificial Intelligence (AAAI) 2025–2026.	
	▪ The European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases (ECML-PKDD) 2026.	
	External Reviewer	
	▪ IEEE Transactions on Knowledge and Data Engineering (TKDE)	
	▪ IEEE Transactions on Games (ToG)	
	▪ ACM Transactions on Intelligent Systems and Technology (TIST)	
AWARDS & HONORS	Outstanding Reviewer, ACM SIGKDD Conference	2025
	▪ Selected as a top 10% reviewer for KDD 2025.	
	Excellence Award, GIST Institute of Artificial Intelligence	2024
	▪ Recognized for outstanding research achievements among postdoctoral researchers.	

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